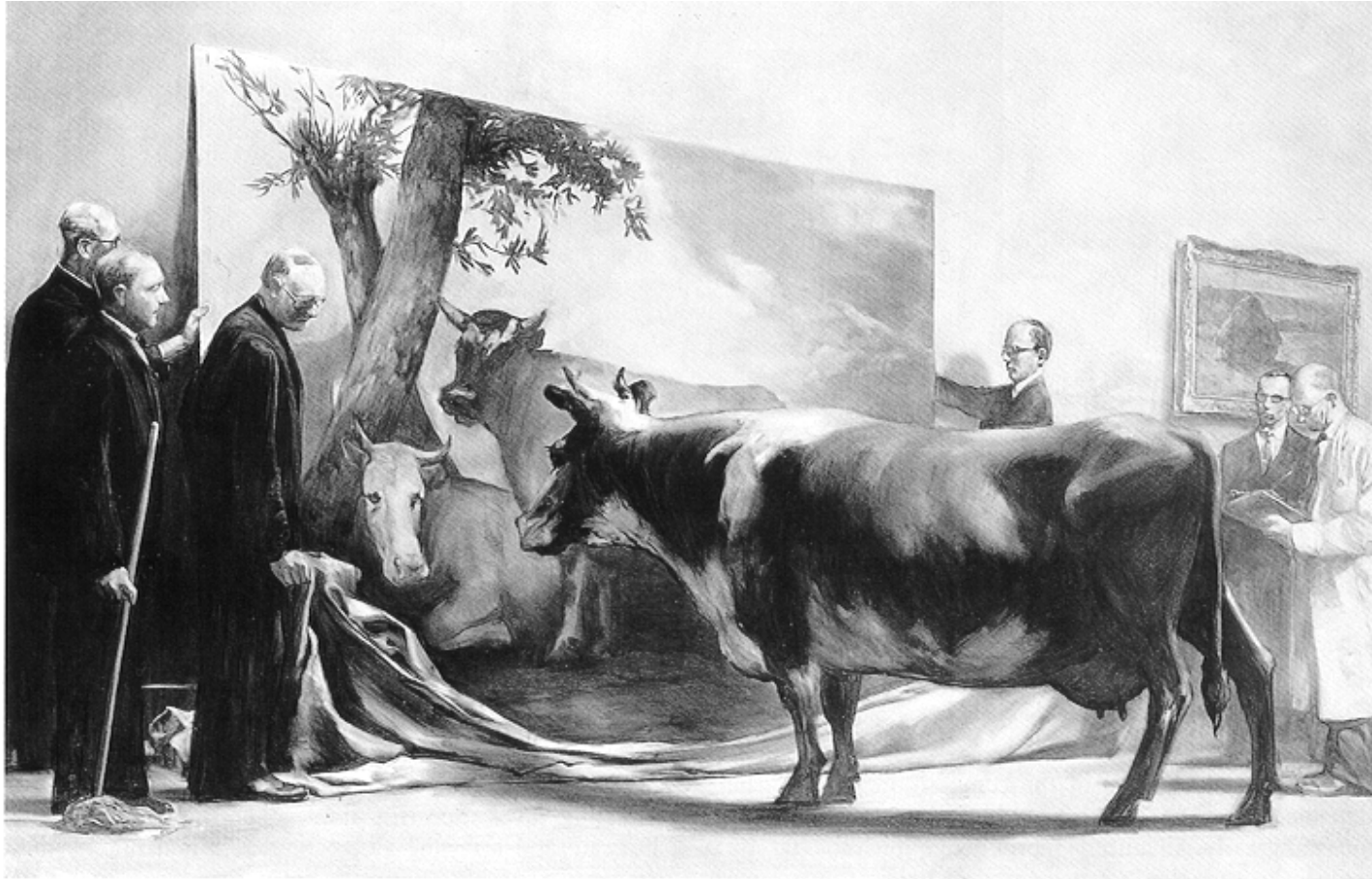


ENVX2001 – Applied Statistical Methods



Lecture 5 – nMDS and hypothesis testing

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Outline

- Multivariate data and
- Hypothesis testing and data exploration*
- Important techniques including
 - PCA
 - Cluster analysis
 - nMDS*
 - Permutational approaches
 - MANOVA

Lectures 3, 4 & 5

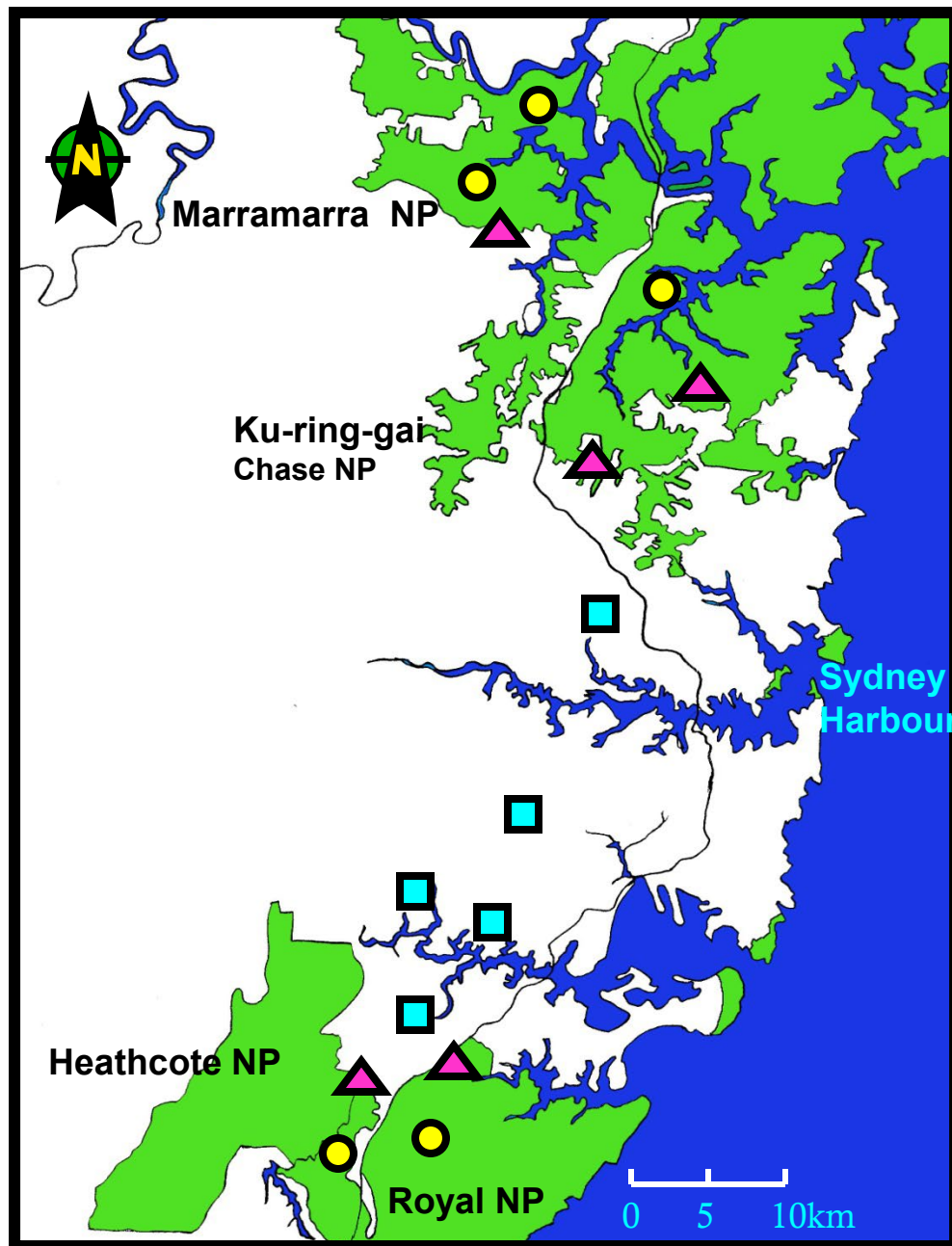
- Distance matrices
- Transformations and standardisations
- Cluster analysis
 - Types of questions
 - Carrying it out
 - Interpreting it
- Hypothesis testing (and best practice)
- Steps for testing
 - Visualising (nMDS)
 - Randomization (generating p using ANOSIM and PERMANOVA)
 - Post-hoc tests
 - Identifying contributors to the differences (SIMPER)

References

- Quinn & Keough (2002) – Chapter 18
- Quinn & Keough (2023) – Chapter 16
- Clarke, K.R. (1993). Non-parametric multivariate analyses of changes in community structure. *Aust. J. Ecol.* **18**:117-43.
- Legendre & Legendre (2012) – Chapter 9
- Somerfield, P.J., *et al.* (2021). Analysis of similarities (ANOSIM) for 2-way layouts using a generalised ANOSIM statistic, with comparative notes on Permutational Multivariate Analysis of Variance (PERMANOVA). *Aust Ecol* **46**:911-26.

nMDS - the question

- Are there differences between ecological communities at different sites?
 - Experimental approaches
 - Comparing landscapes
- Confidence that they are different?
- Bad experiments = bad conclusions
 - Spatial independence of sites
 - undersampling



- Legend**
- Interior sites
 - Edge sites
 - Small sites

Cluster analysis

Hypothesis testing

nMDS

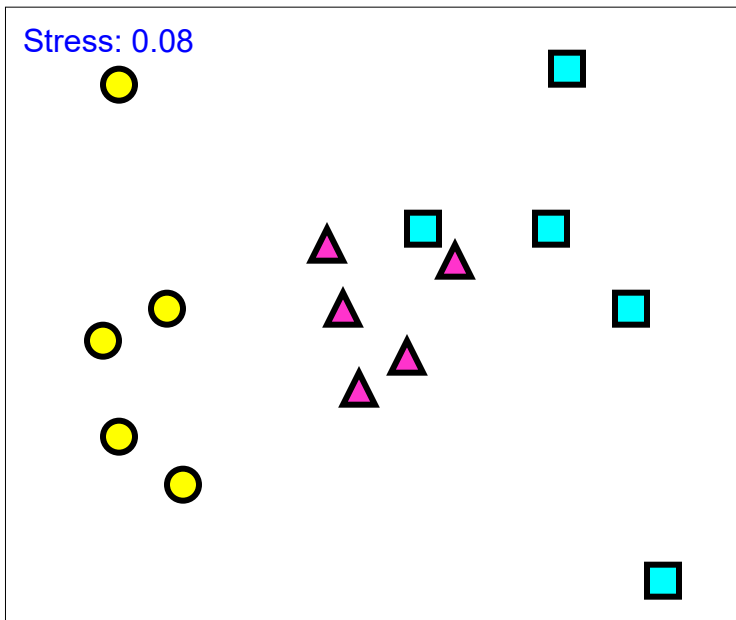
ANOSIM

SIMPER

nMDS - the output

Global R = 0.624; P < 0.001

Stress: 0.08



**Bird composition is
different for all sites
(Within & Inside)
(Presence/Absence)**

-  Small
-  Edge
-  Interior



Distance between points reflects relative similarity

nMDS - the process/meaning

- Generate distance matrix
- nMDS
 - stress
- ANOSIM
 - Global R
 - Post hoc tests
- PERMANOVA as an option*** (Tomorrow)
- SIMPER
 - Which species are contributing the differences

A process

- Collection of data and synthesis into matrix
- Measures of similarity and similarity matrix
- Ordinate (using nMDS) or cluster
- ANOSIM/PERMANOVA
 - Experiment wide
 - Post hoc tests
- SIMPER

Ordination (scaling)

- Mapping in multidimensional space to reveal patterns
- Of the methods, some;
- Require specific distance measures (euclidean in PCA, RDA, chi-square for CA & CCA) not ideal for species abundance data
- Have built-in standardisations (e.g. CA)
- assume linear relationships (RDA, CCA)

Nonmetric multidimensional scaling (nMDS)

“the future of ordination is in nonmetric multidimensional scaling” – McCune & Grace, 2002

Nonmetric: no axes, not euclidean

Multidimensional: represents relationships between multiple variables in two or three dimensions

Scaling: the ratio between reality and representation

nMDS

- Non-metric multi-dimensional scaling
- Uses rank orders
- Robust – not laden with assumptions
- Amenable to:
 - Various standardisations and transformations
 - Flexible with respect to dissimilarity measures
 - Useful for describing patterns and testing *a priori* hypotheses
- Vegan – R package routine for running these tests

How does nMDS work?

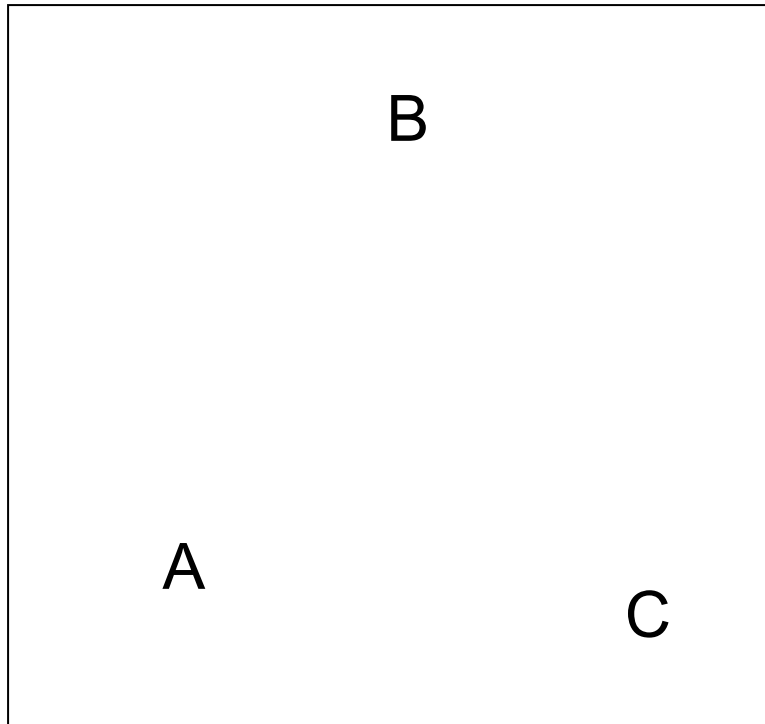
nMDS uses the RANK ORDER of similarity relationships between samples:

Sample	Sample	% similarity	rank
A	B	99%	1
A	C	96%	2
B	C	95%	3

A is closer to B than it is to C

How does nMDS work?

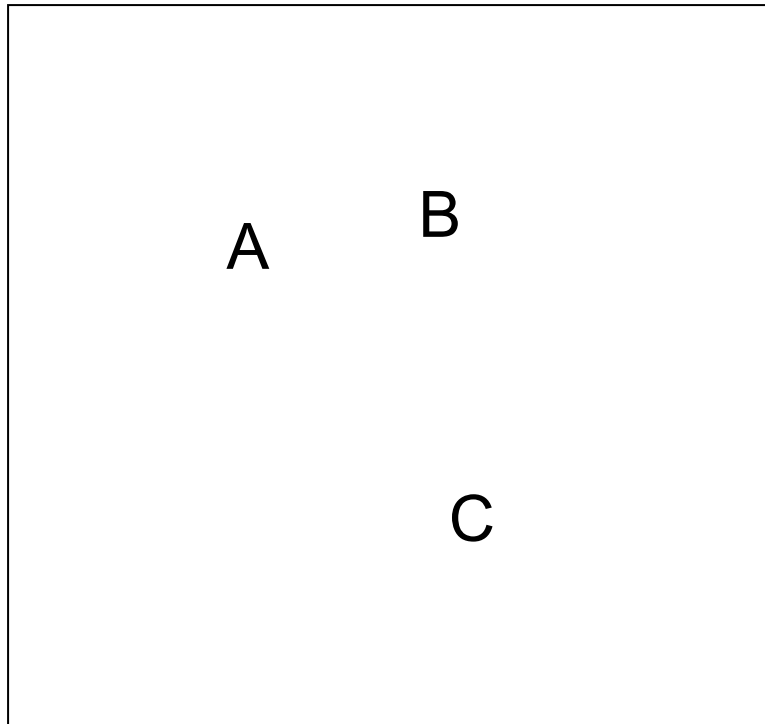
Then, nMDS tries to place points in 2 (or 3) dimensional space to represent this ranked order:



A is closer to B
than it is to C in
the similarity
matrix

How does nMDS work?

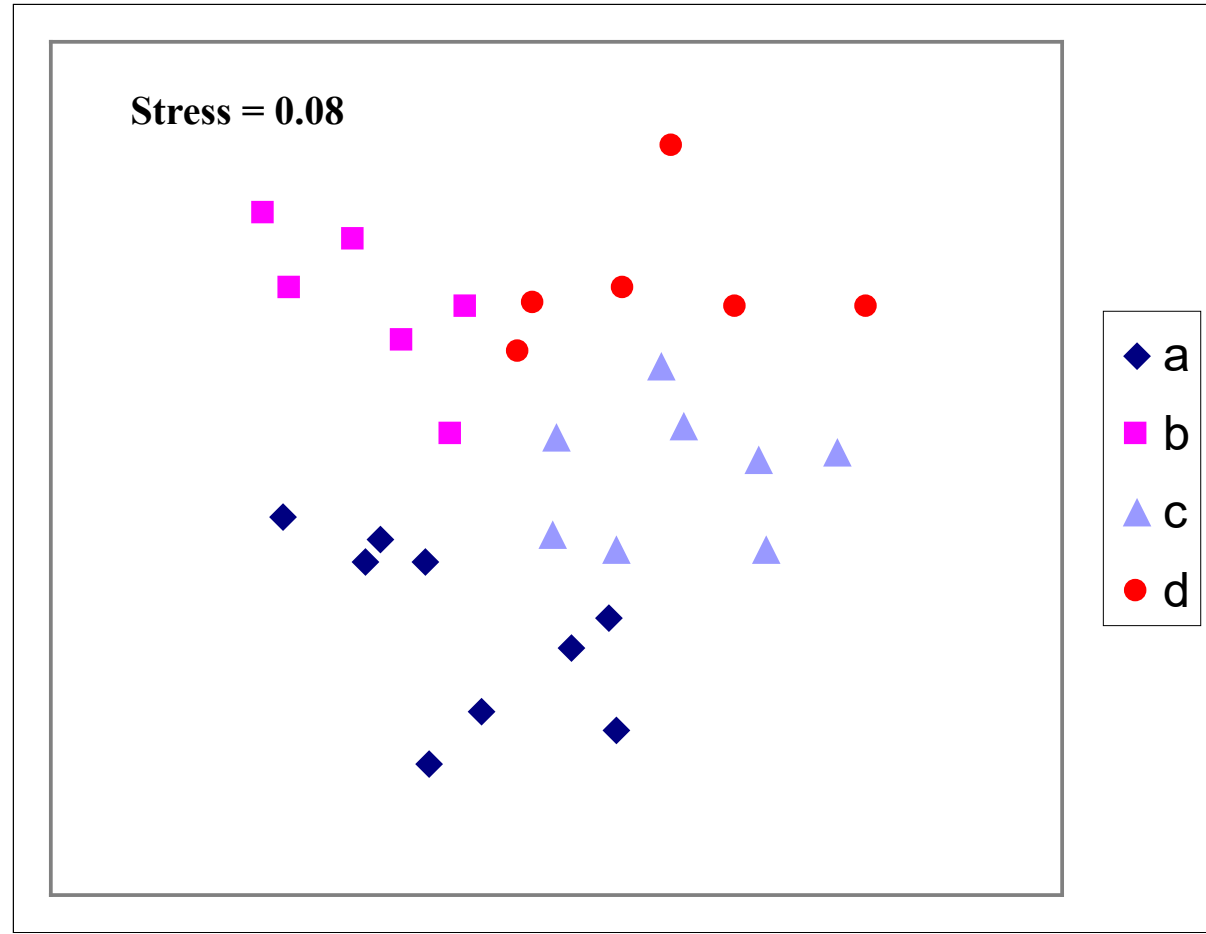
Then, nMDS tries to place points in 2 (or 3) dimensional space to represent this ranked order:



A is closer to B
than it is to C

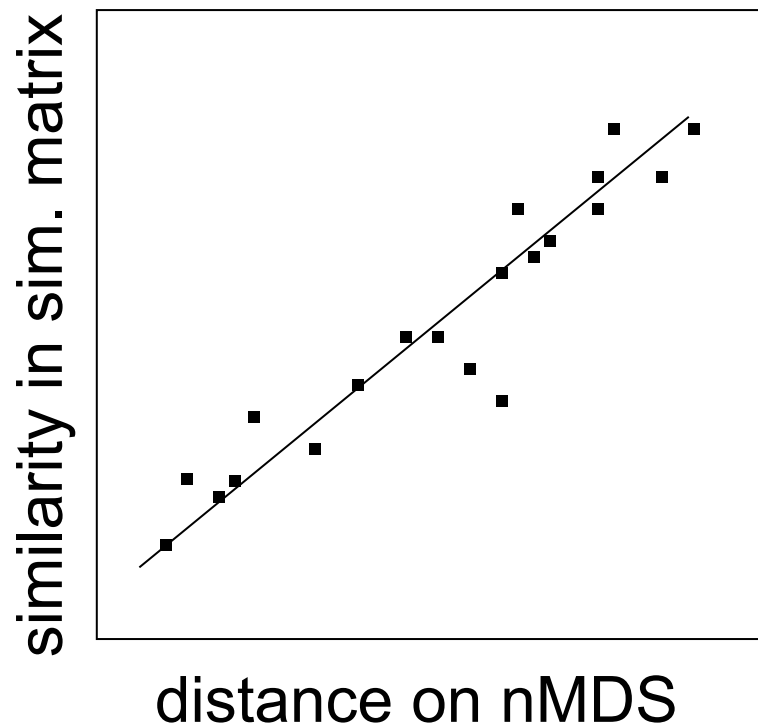
How accurate is the nMDS map?

Sometimes nMDS
can't represent all
relationship
accurately



How accurate is the nMDS map?

- Sometimes the nMDS can't represent all relationship accurately
- this is reflected by a high STRESS value

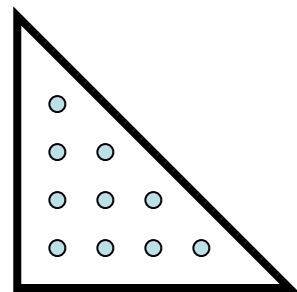


If Stress Value =

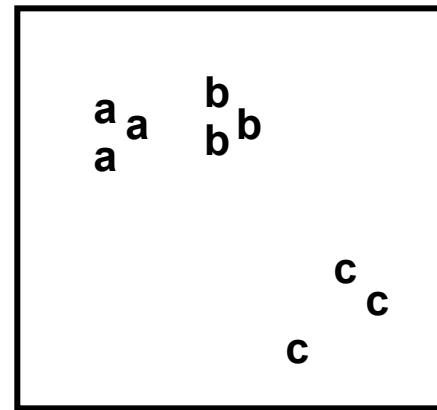
- 0.0 : perfect map
- 0.1 : decent map
- 0.2 : ok map
- 0.3 : don't bother

Main points about ordination

- Ordination is a way to visualize how similar samples are
- nMDS tries to represent visually the rank order within the underlying similarity matrix
- all that matters is the relative distance between points.
- stress value allows you to estimate 'quality' of the nMDS'



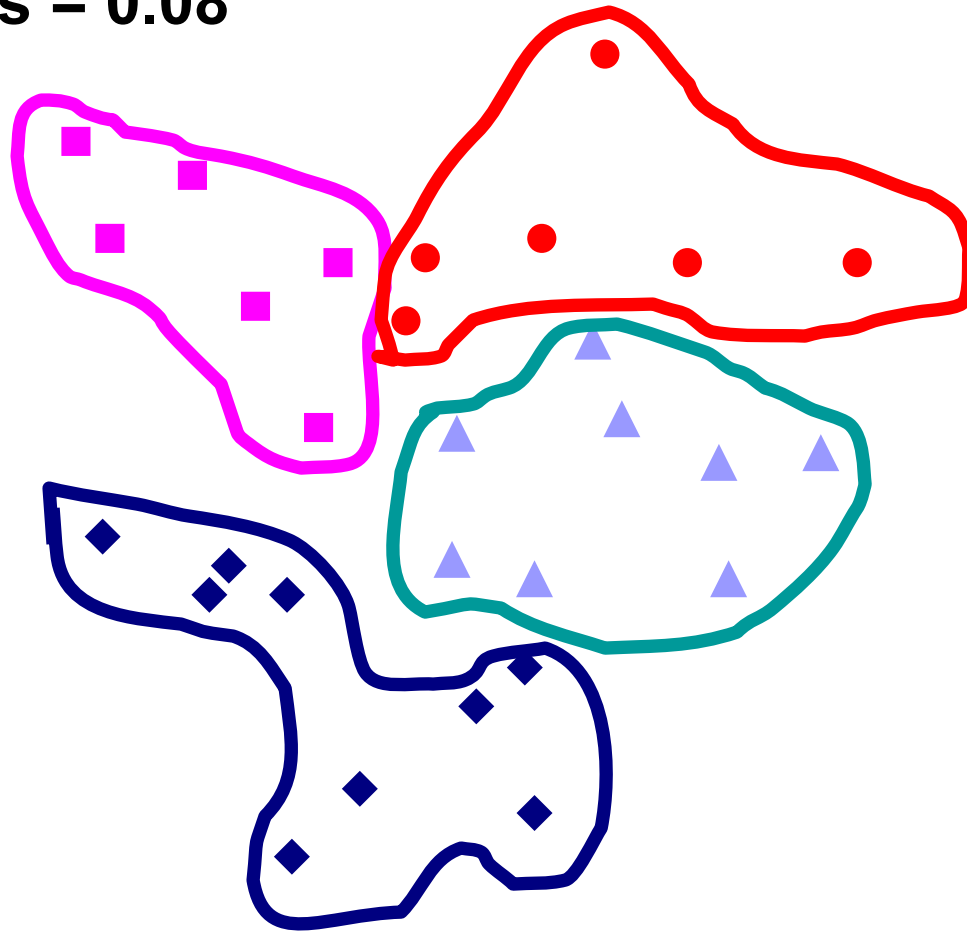
sample
similarities



ordination

Interpretations?

Stress = 0.08



- ◆ a
- b
- ▲ c
- d

How to do a nMDS in R

```
# Many ways -> this is using vegan package
#Need a dissimilarity matrix first
copepod.dis <- vegdist(TransCopepodData,
method="bray")

# Then you can plot the NMDS using the
metaMDS command

pchs<- c(0:2)

Copepod_Factor <-
factor(CopepodData$Treatment)

nMDS_copepod <- metaMDS(TransCopepodData,
distance="bray", k=2)
```

How to do a nMDS in R (cont.)

```
# Here we actually plot it
plot_copepod <- ordiplot(nMDS_copepod,
display = "sites", type="n")
# Add the points
points(nMDS_copepod, col="black", pch =
pch$Copepod_Factor)
# Add the legend
legend("topright", bty = "n", legend =
levels(Copepod_Factor), pch = pch$Copepod_Factor)
```

Permutational tests

- Assumptions relaxed c.f. parametric stats
- Example focuses on ANOSIM
- PERMANOVA a good option when you have a more complex design



ANOSIM (Clarke & Warwick 1994)

- Analysis of similarities
- Retains philosophy underpinning nMDS, driven by hypothesis testing
- Handles simple designs best (1 factor)
- Non-parametric permutation test combined with randomization approach to generate significance levels (Monte Carlo tests)
- Null – no difference in assemblage composition among sites

ANOSIM

- Null - Average rank dissimilarities between all possible pairs of samples in different groups is the same as average of the rank dissimilarities between pairs of samples within groups
- analogous to ANOVA in that it compares ranks among groups to ranks within group
- Global test followed by pairwise comparisons
- Limitation – complex tests and interactions are difficult to handle (but see Anderson 2001 for options)



Testing hypotheses

$$\text{Test statistic} = \frac{\text{Difference between things}}{\text{Difference within things}}$$

Testing hypotheses

$$\text{Test statistic} = \frac{\text{Difference between things}}{\text{How much they each vary}}$$

Testing hypotheses

$$\text{e.g. } t = \frac{\bar{X}_1 - \bar{X}_2}{(s.e._1 + s.e._2)/2}$$

Are groups different?

Analysis of Similarities – a statistical approach

$$R = \frac{(r_{\text{between}} - r_{\text{within}})}{\text{standardizing factor}}$$

From dissimilarity matrix to hypothesis test

Sample	1	2	3	4
1	-			
2	25.6	-		
3	0.0	67.9	-	
4	52.2	68.1	42	-

Nb. ANOSIM needs more samples ($n > 4/\text{group}$) to complete randomisation tests and test at 0.05



Exposed and sheltered rocky shores

Source: The Field Studies Council (FSC) , UK

Effect of Wave Energy on Plant Herbivore Interactions



Protected

Exposed



Ecklonia radiata

Turbo torquatus

Bray-Curtis Similarity Matrix

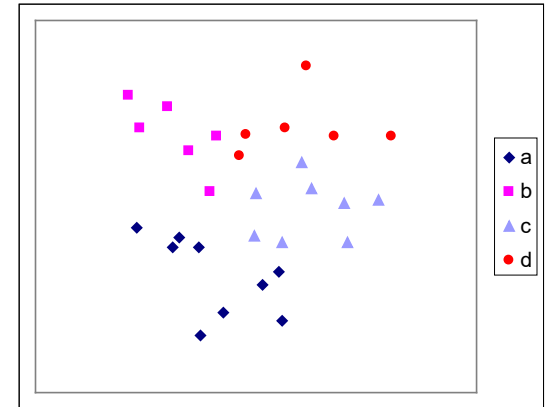
	exposed	exposed	exposed	exposed	exposed	exposed	exposed	exposed	exposed	sheltered	sheltered	sheltered	sheltered	sheltered	sheltered	sheltered	sheltered
exposed																	
exposed	86.816																
exposed	86.434	81.454															
exposed	27.454	72.823	48.918														
exposed	76.357	42.598	65.521	31.3													
exposed	34.711	32.506	49.061	50.639	47.66												
exposed	36.569	59.381	48.367	60.333	24.247	23.824											
exposed	46.486	10.268	40.421	7.7514	44.605	33.767	9.156										
sheltered	30.977	20.188	26.991	15.303	32.534	55.651	9.007	43.116									
sheltered	52.942	41.312	47.162	32.754	44.165	39.942	22.559	36.648	44.685								
sheltered	50.599	40.918	43.643	30.384	59.581	59.275	9.6934	48.333	70.863	43.965							
sheltered	49.376	18.166	43.546	14.113	39.588	47.957	8.1935	51.435	65.745	68.965	60.525						
sheltered	0	15.145	0	10.24	0	14.326	0	0	33.997	26.701	13.793	37.361					
sheltered	0	38.191	0	23.812	0	15.879	15.546	0	14.992	22.558	16.995	12.866	29.709				
sheltered	0	11.251	0	8.2986	0	35.342	0	13.929	62.723	21.827	43.653	51.33	67.436	17.696			
sheltered	15.151	28.299	12.956	20.641	16.031	30.604	0	14.432	51.995	30.683	52.975	43.937	41.001	18.878	57.455		

Within

Within

Between

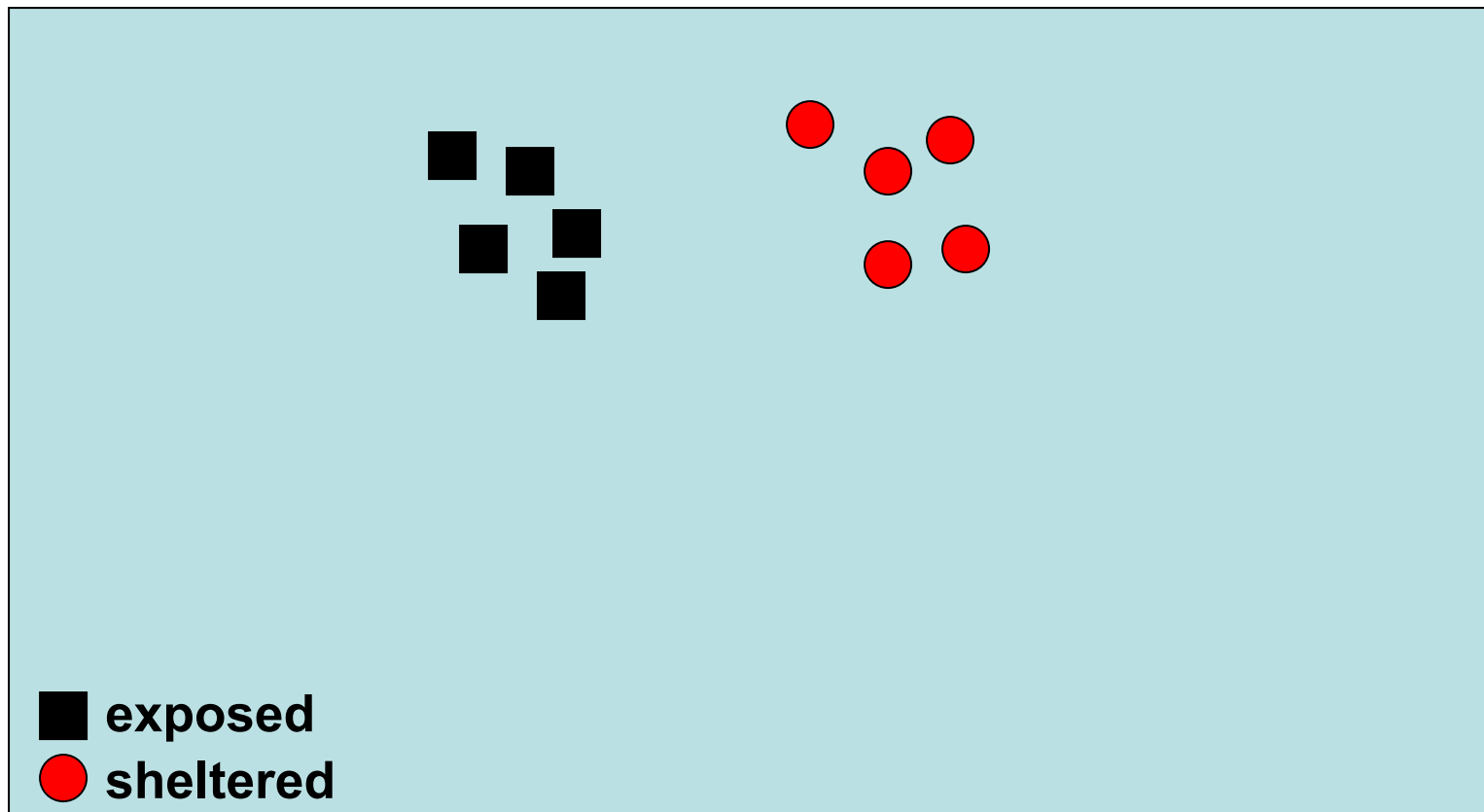
Steps for ANOSIM



1. Test statistic – reflects difference among sites to differences within sites. Based on rank similarities in the underlying triangular matrix, generates “Global R”
2. Recompute R under permutations of sample labels. If null is true there should be little change to R if labels arbitrarily assigned.
3. Significance value – refer the observed value of R to permutation distribution. If it is unlikely (<5%) to fall in this distribution, reject H_0

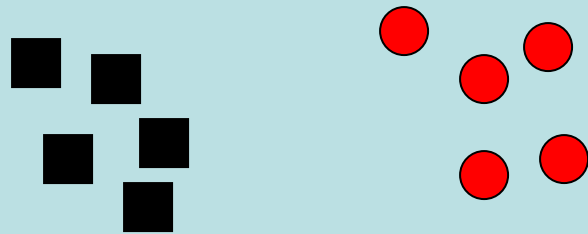
Are groups different?

Analysis of Similarities – a statistical approach



Are groups different?

Analysis of Similarities – a statistical approach

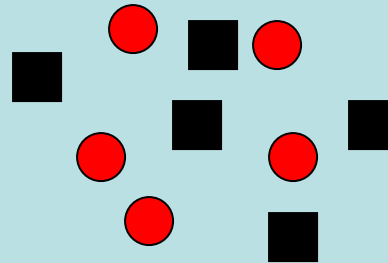


H_o = sites the same

H_a = sites are different

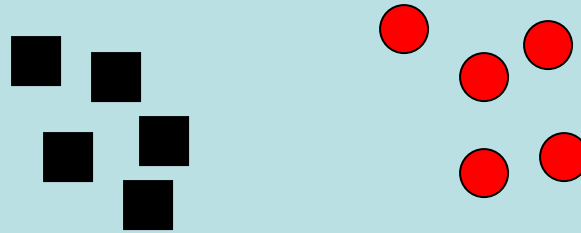
■ exposed
● sheltered

If H_0 (sites the same) = true



Similarity within = Similarity between

If H_a (sites different) = true



Dissimilarity between > Dissimilarity within

Are groups different?

Analysis of Similarities – a statistical approach

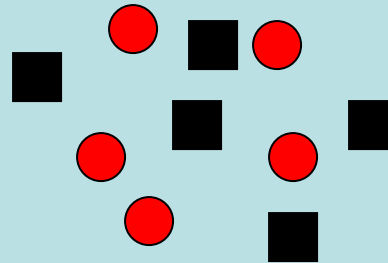
$$R = \frac{(r_{\text{between}} - r_{\text{within}})}{\text{standardizing factor}}$$

Are groups different?

Analysis of Similarities – a statistical approach

$$R = \frac{(r_{\text{between}} - r_{\text{within}})}{\sim 1}$$

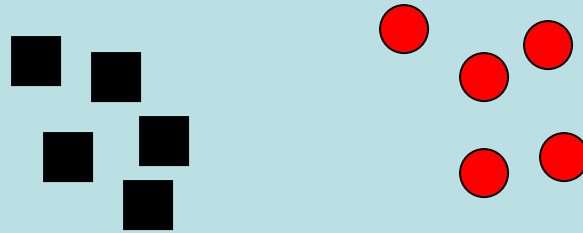
If H_0 (sites the same) = true



Dissimilarity within = Dissimilarity
between

$$R = \frac{(r_{\text{between}} - r_{\text{within}})}{\sim 1} \sim 0$$

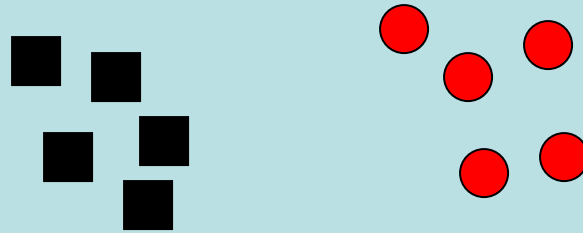
If H_a (sites different) = true



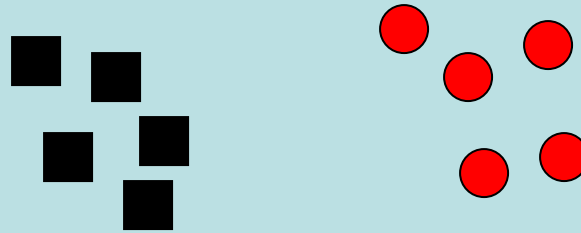
Dissimilarity between > Dissimilarity within

$$R = \frac{(r_{\text{between}} - r_{\text{within}})}{\sim 1} \sim 1$$

To simulate null distribution

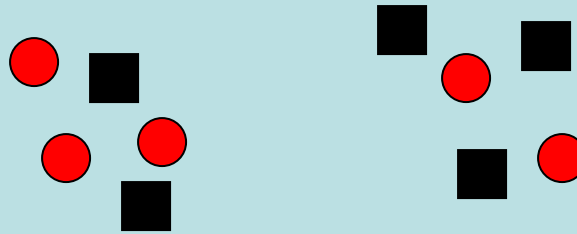


To simulate null distribution



**Dissimilarity within = Dissimilarity
between**

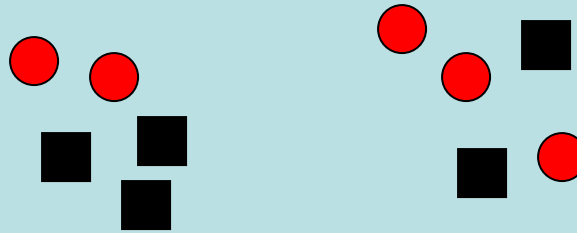
To simulate null distribution



Variation within = Variation between

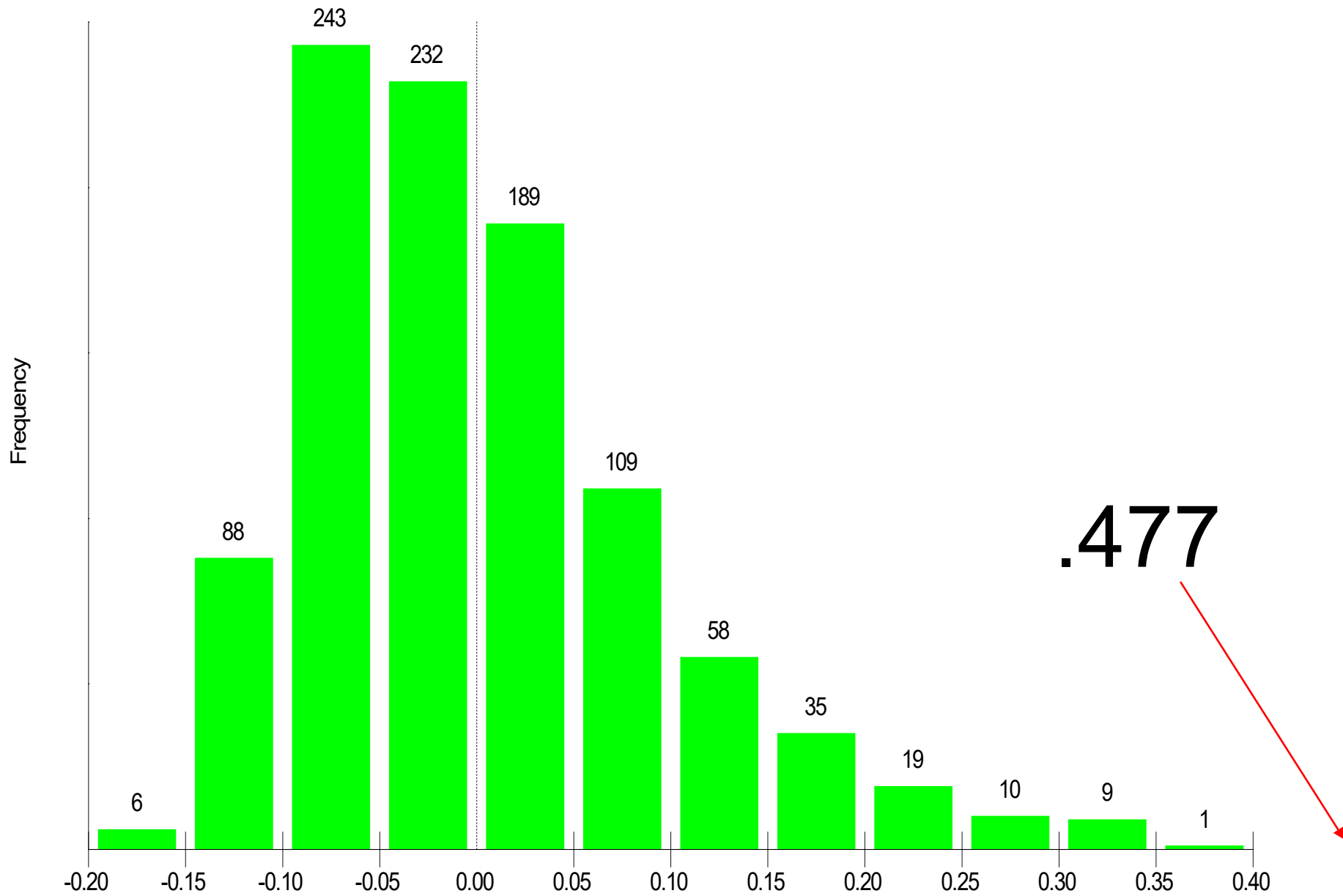
Calculate R

To simulate null distribution



**Dissimilarity within = Dissimilarity
between**

Calculate R



.477

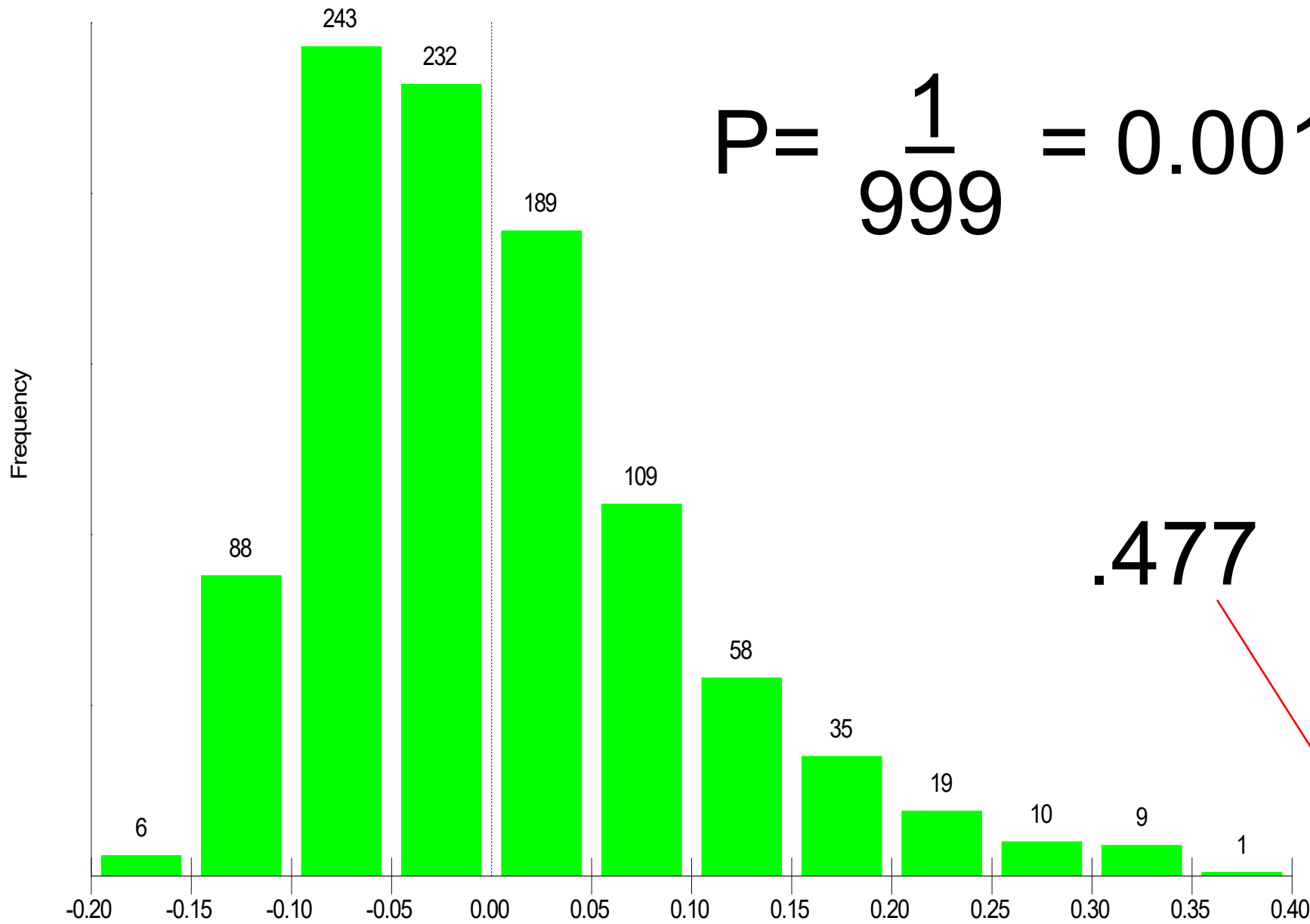
Cluster analysis

Hypothesis testing

nMDS

ANOSIM

SIMPER



Cluster analysis

Hypothesis testing

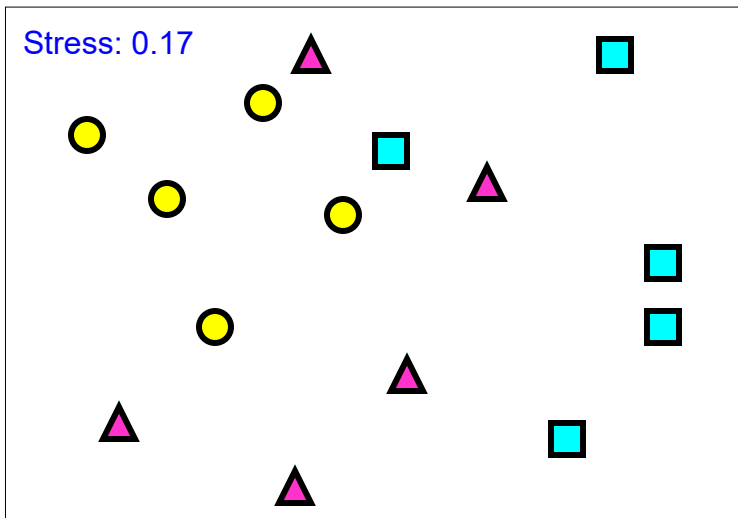
nMDS

ANOSIM

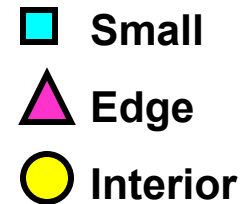
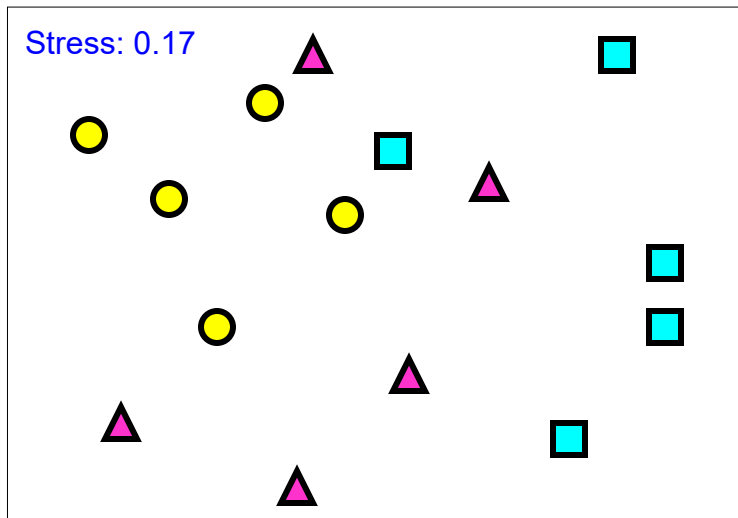
SIMPER

Is bird composition different at small, edge & and interior sites? (Within) (4th root)

- Small
- ▲ Edge
- Interior



Is bird composition different at small, edge & and interior sites? (Within) (4th root)



ANOSIM statistic (Global R): 0.42

Significance: 0.003

Number of permutations: 999 (Random sample from 126126)

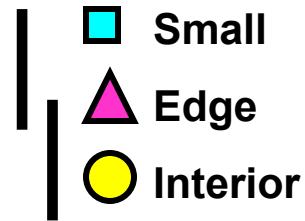
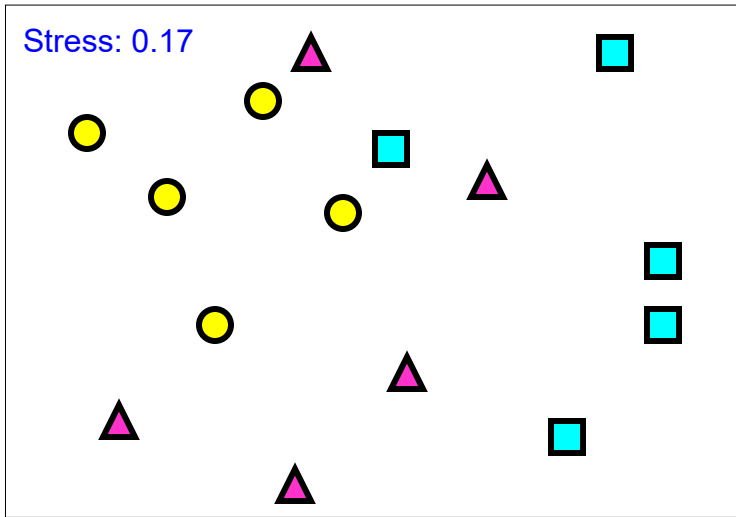
Number of permuted statistics greater than or equal to Global R: 2

Pairwise Tests

Groups	R Statistic	Significance Level	Possible Permutations	Actual Permutations	Number >= Observed
E, I	0.256	0.07	126	126	9
E, S	0.132	0.19	126	126	24
I, S	0.768	0.008	126	126	1

Global R = 0.42; P = 0.003

Is bird composition different at small, edge & and interior sites? (Within) (4th root)



ANOSIM statistic (Global R): 0.42

Significance: 0.003

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Pairwise Tests

Groups	Statistic	R	Significance Level	Possible Permutations	Actual Permutations	Number >= Observed
E, I	0.256		0.07	126	126	9
E, S	0.132		0.19	126	126	24
I, S	0.768		0.008	126	126	1

How to do an ANOSIM in R?

```
library(vegan)
#Need to generate dissimilarity first
copepod.dis <- vegdist(TransCopepodData,
method="bray")
copepod.anosim <- with(CopepodTreatment,
anosim(copepod.dis, Treatment))
summary(copepod.anosim)
#This is the anosim command in the package
vegan
```

Sites are different – why?

- The SIMPER routine:
 - Similarity Percentages
- Which variable contribute most to the observed patterns?
- For ecological data – which species contribute most to the separation among sampling units?

SIMPER

- Calculates % contribution of each species to the dissimilarities between all pairs of sampling units in different groups and % contribution to similarities between all pairs within groups
- Output- a list of species in order of their percent contributions to dissimilarities between or similarities within groups
- Note some criticism of use – see Warton, et al.. 2012. Method Ecol Evol 3: 89–101.

Parameters
Standardise data: No
Transform: Fourth root
Cut off for low contributions: 80.00%
Factor name: Treatment
Factor groups

E
I
S

Group E

Average similarity: 33.41

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
RALO	9.20	5.74	1.10	17.19	17.19
EASP	2.20	4.53	1.15	13.57	30.76
GRFT	1.20	3.78	1.13	11.32	42.08
EYRO	1.20	2.03	0.62	6.07	48.15
SPPA	2.60	1.94	0.62	5.80	53.96
BRTB	4.00	1.88	0.61	5.62	59.58
CRRO	1.80	1.88	0.61	5.62	65.20
LIWB	1.20	1.80	0.62	5.39	70.59
LAKO	1.00	1.75	0.62	5.23	75.82
AURA	1.20	1.74	0.62	5.20	81.02

Group I

Average similarity: 46.01

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
BRTB	10.00	8.13	12.49	17.68	17.68
EYRO	2.80	6.30	3.75	13.68	31.36
GRFT	2.60	5.93	12.39	12.89	44.25
WBSW	2.80	3.98	1.12	8.64	52.90
YFHE	2.60	3.47	1.15	7.54	60.44
CRRO	1.40	3.46	1.14	7.52	67.96
GRST	1.20	3.18	1.12	6.91	74.87
EASP	2.80	2.96	1.10	6.43	81.30

Group S

Average similarity: 39.59

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
RALO	16.80	11.21	4.61	28.32	28.32
CRRO	5.20	5.49	1.16	13.86	42.18
NOMI	7.80	3.14	0.62	7.94	50.12
SILV	7.60	2.94	0.62	7.43	57.55
VAFW	1.60	2.63	0.62	6.64	64.19
AURA	2.80	2.61	0.62	6.59	70.78
WBSW	1.80	2.50	0.61	6.32	77.10
PICU	1.40	2.30	0.62	5.80	82.91

- Which species help discriminate?



Neil Fifer Sydney,Aust Jul 2004

Groups E & I
Average dissimilarity = 65.41
...snip...

Groups E & S
Average dissimilarity = 66.19
...snip...

Groups I & S
Average dissimilarity = 76.43
Group I Group S



Species	average	sd	ratio	ava	avb	cumsum
RALO	5.75	1.737	3.31	0	16.8	7.52
BRTB	4.22	1.758	2.4	10	1	13.04
GRFT	3.64	0.302	12.06	2.6	0	17.8
NOMI	3.25	2.928	1.11	0	7.8	22.05
SILV	3.05	2.723	1.12	1.8	7.6	26.05
SCCO	2.89	3.108	0.93	16.6	5.2	29.83
YFHE	2.64	1.66	1.59	2.6	0.2	33.29
EYRO	2.6	1.831	1.42	2.8	0.6	36.68
AURA	2.56	2.226	1.15	0	2.8	40.04
EASP	2.27	1.802	1.26	2.8	0.4	43.01
GRST	2.23	1.604	1.39	1.2	0.2	45.93
CRRO	2.05	1.414	1.45	1.4	5.2	48.62

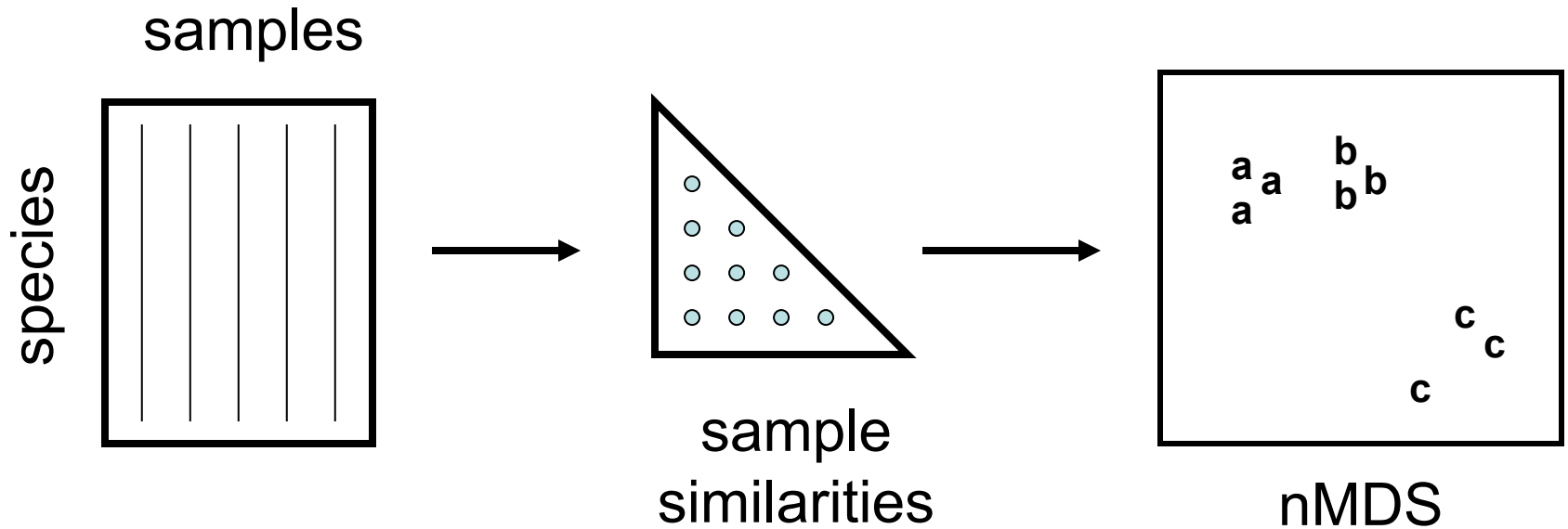
How to interpret the SIMPER table?

- average = the average dissimilarity between treatments, contributes to overall dissimilarity
- sd = standard deviation of the dissimilarity which is the variation in the differences between treatments
- ratio = average / sd which is the dissimilarity taking account of the variation
- ava and avb = the average number of that species in each treatments
- cumsum = the ordered cumulative contribution of that species to the overall dissimilarity – useful as a guide for deciding how many species are useful

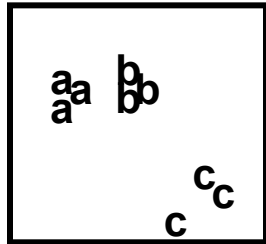
How to do a SIMPER in R?

```
copepod.simper <-  
simper(CopepodData,  
CopepodTreatment$Treatment)  
summary(copepod.simper)  
#Again this is in the vegan  
package
```

Data analysis summary

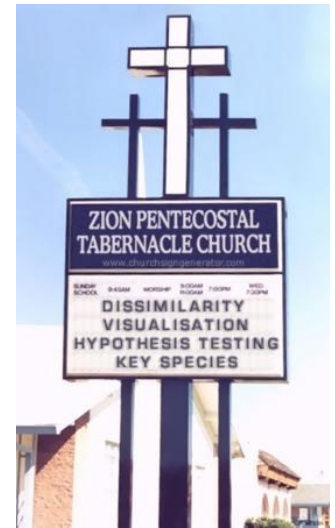


are sites different?



ANOSIM/PERMANOVA

How?
SIMPER



nMDS and ANOSIM - summary

- Robust ordination technique (few assumptions)
- Hypothesis test consistent with ordination
- Clarke (1993) makes strong case for it as the best approach for dealing with species abundance data
- Limits of ANOSIM – need adequate replication for permutation tests
- PERMANOVA as an option for more complex designs – tomorrow's lecture

Outline

- Multivariate data and extending ecology
- Hypothesis testing and data exploration*
- Important techniques including
 - PCA
 - Cluster analysis
 - nMDS
 - Monte Carlo approaches
 - **MANOVA/PERMANOVA**